

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Sixth Semester of B. Pharm. Examination
University Theory Examination September 2018
PH325 Pharmaceutical Technology-I

Date: 18/09/2018, Tuesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION – I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. Wurster process for fluidized bed coating is also known as _____.
[A] Tangential Spray
[B] Top Spray
[C] Bottom Spray
[D] Side Spray
2. Ethyl Cellulose is an example of _____.
[A] Water soluble resin
[B] Waxes and lipids
[C] Enteric resins
[D] Water Insoluble resins
3. SEM is a technique used for determination of _____.
[A] Particle size and shape
[B] Density
[C] Isoelectric point
[D] Angle of Repose
4. _____ is an example of oleaginous base.
[A] PEG
[B] Hydrophilic ointment
[C] Lanolin
[D] Yellow soft paraffin
5. Creams are generally prepared by _____.
[A] Lavigation method
[B] Fusion method
[C] Trituration method
[D] Tile method

6. _____ size of particles penetrate follicles and stratum corneum alike.
[A] Greater than 10 micrometer
[B] Between 3 and 10 micrometer
[C] Less than 03 micrometer
[D] Less than 10 micrometer
7. Implementation of GMP is an investment in _____.
[A] Spurious medicines
[B] Adulterated medicines
[C] Good quality medicines
[D] Bad quality medicines
8. A _____ should be prepared, summarizing the results obtained, commenting on any deviations observed, and drawing the appropriate conclusions, including recommending changes to correct deficiencies.
[A] Validation report
[B] Validation policy
[C] Validation protocol
[D] Validation System
9. _____ is an example of biodegradable polymer.
[A] Polyanhydride
[B] PMMA
[C] Acrolein
[D] Epoxy polymers
10. Commonly used diluent in preparation of pellets is _____.
[A] MCC
[B] Fructose
[C] Mannitol
[D] MgO
11. Which of the following one is used as base for chewable tablets?
[A] Fructose
[B] Glucose
[C] Lactose
[D] Mannitol
12. Which of the following one is used as opacifier?
[A] Oxalates
[B] MgO
[C] Silicates
[D] TiO₂
13. Capping is prevented by using _____ punches.
[A] Rectangular
[B] Circular
[C] Flat
[D] Square

14. In the tablet coating process, inadequate spreading of coating solution before drying is known as _____.
 [A] Picking effect
 [B] Sticking effect
 [C] Blistering effect
 [D] Orange peel effect
15. Diclofenac tablet coated with CAP has been administered to a patient. Where do you expect the drug to be released?
 [A] Stomach
 [B] Small Intestine
 [C] Large Intestine
 [D] Oral Cavity
16. Shellac is used in the coating of tablets as _____.
 [A] Enteric coating agent
 [B] Film coating agent
 [C] Sub-coating agent for sugar coating
 [D] Polishing Agent
17. Which of the following is a water soluble lubricant?
 [A] Stearic acid
 [B] Mineral oil
 [C] PEG
 [D] Magnesium stearate
18. Unequal distribution of color on a tablet, refers to _____.
 [A] Picking
 [B] Mottling
 [C] Capping
 [D] Sticking
19. Starch, on heating, hydrolyze into _____.
 [A] Dextrin & Glucose
 [B] Fructose & Sorbose
 [C] Fructose & Mannose
 [D] Glucose
20. Documented verification that the equipment or systems, as installed or modified, perform as intended throughout the anticipated operating ranges is known as _____.
 [A] DQ
 [B] OQ
 [C] PQ
 [D] IQ

SECTION – II

Q 2 Attempt any **TWO** of the following

- | | | |
|----------|---|-----------|
| A | Describe the tablet defects and their remedies. | 05 |
| B | Explain the ideal requirements of the ointment bases. | 05 |

- C** Enlist the needs for the formulation of pellets. Write a note on extrusion spheronization process for the formulation of pellets. **05**
- Q 3** Attempt any **TWO** of the following
- A** Construct a manufacturing flow sheet for production of Cetrizine fast disintegrating tablet. Suggest the suitable ingredients for the fast dissolving tablet. **05**
- B** Justify the need of climate control in capsules formulation departments. List down the different size of hard gelatin capsules. **05**
- C** Describe the advancement in packaging technology for pharmaceutical products. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Write a note on sugar coating of tablet. **05**
- B** Explain, with suitable diagram, the rotary die method for the formulation of soft gelatin capsules. **05**
- C** Explain in detail a microencapsulation process in which dispersion of core particle in a polymer solution is sprayed into hot chamber with a well labeled diagram. **05**
- D** Discuss the various factors affecting release of drug from semisolids in brief. **05**
- E** Write a note on ideal requirements of materials used in packaging of pharmaceutical products. **05**
- F** Describe tamper proof packaging systems for pharmaceuticals. **05**
- Q 5** Attempt any **FOUR** of the following
- A** Describe the evaluation parameters for calcium lactate chewable tablets. **05**
- B** List down the objectives for coating of tablets. Enumerate the enteric film formers. **05**
- C** Describe the manufacturing process of empty hard gelatin capsules. **05**
- D** Enlist ideal properties, advantages and disadvantages of microcapsules. **05**
- E** Write a note on: Evaluation of ointment. **05**
- F** Discuss in brief the good manufacturing practice for pharmaceuticals. **05**